Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Based Co-Design Interface for Robotic Systems			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics / Automation Machinery	Robot Design Technologies	*Human Resource Competency Development Services*	Human–System Interaction Technologies
Concept of Proposed Technology	The proposed technology aims to develop an AI-based co-design interface for robotic systems that allows users to design, modify, and optimize robots and mechanisms through intuitive inputs such as sketches and language. By combining artificial intelligence with mathematical optimization, the system supports both rapid concept generation and fine-tuning of designs, while enabling a human-centered co-design process that provides an effective and accessible user experience. ● Sketch- and text-based design input that enables intuitive robot and mechanism creation			

	• AI-assisted design generation and refinement, supported by optimization-based performance tuning • Human-centered co-design workflow that adapts designs to individual users and design intentions
Rationale for Joint R&D Project	This project requires close collaboration between a research institute and an industrial partner to successfully realize an AI-based co-design interface for robotic systems that is both technically advanced and practically usable for end-users, such as robot designers. **Rationale for collaboration:** • Georgia Tech brings strong expertise in AI algorithms, optimization, HCI, and user-centered interface design. • The company brings strengths in robot hardware, system implementation, and practical design constraints. • Effective co-design of algorithms, interfaces, and real robotic systems is essential to ensure realism and applicability. **Expected benefits of joint R&D:** • Faster transition from research prototypes to deployable design tools. • Validation of the interface through real hardware and realistic design scenarios • Creation of a platform that is both technically powerful and user-friendly, suitable for commercialization • Strong synergy between AI research, HCI/user studies, and industrial robotics development
Key Words	1. AI-Assisted Robot Design 2. Human–AI Co-Design Interface 3. User-Centered Robot Design (HCI)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The goal of this project is to create an AI-based co-design platform for robotic systems that allows users to intuitively design and refine robots using intuitive interfaces, such as sketches or language inputs. It also aims to bridge the gap between conceptual design and real-world robotic performance. Objectives: ● To build an interactive AI-assisted design interface that supports sketch- and text-based inputs for rapid robot and mechanism creation. ● To develop a user-centered co-design workflow and validate its effectiveness through usability studies and human-centered evaluation. ● To ensure a high-fidelity connection between design and execution.			
R&D Target Specifications	● **Usability Performance (>75):** Achieve a System Usability Score (SUS) greater than 75, validated through structured user studies with representative users. ● **Physical Accuracy Performance (<1cm):** Achieve an end-effector control error less than 1 cm when robots designed through the interface are implemented on real hardware.			
R&D Plan	● **Year 1:** Problem Definition and Initial Interface Development ○ Define target robot types and design scenarios with the company ○ Develop an initial AI-assisted design interface ○ Implement basic sketch/text-based input and design generation functions ● **Year 2:** Interface Enhancement and AI Performance Improvement ○ Refine the interface based on user feedback and early user studies ○ Improve AI models and optimization methods for higher-quality designs ○ Strengthen the connection between design output and robotic performance ● **Year 3:** Experimental Validation and Commercialization Preparation ○ Conduct full user studies and hardware experiments ○ Validate usability (SUS) and physical accuracy (end-effector error) ○ Finalize the system for practical deployment and commercialization			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (Georgia Tech) will be responsible for the development of the core AI algorithms, optimization methods, and user-centered design framework for the proposed co-design interface. The expected responsibilities include: ● Development of AI models and optimization algorithms for robot and mechanism design and fine-tuning. ● Definition of design scenarios and workflows in collaboration with the company. ● Planning and execution of user studies and usability evaluations (e.g., SUS measurement). ● Technical leadership in system architecture and overall algorithmic integration.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the practical realization, system integration, and experimental validation of the proposed AI-based co-design interface in real robotic systems. The expected responsibilities include: ● Definition and refinement of practical design scenarios and target robot types based on industrial needs. ● Integration of the developed system with actual robotic platforms and control systems. ● Execution of hardware experiments and performance evaluations, including end-effector accuracy testing. ● Contribution to technology refinement and commercialization planning, leveraging industrial experience and market insight.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project is expected to deliver a validated AI-based co-design interface that enables intuitive, efficient, and accurate robot design through human–AI collaboration. The resulting system will demonstrate that complex robotic systems can be designed and optimized by users using sketch- and text-based interaction while maintaining high physical accuracy. From an industrial perspective, the project will provide a practical design platform that reduces development time and cost for robotic systems, and improves the usability and accessibility of robot design tools. It will strengthen the competitiveness of the Korean SME partner by enabling faster prototyping, better customization, and more efficient product development. More broadly, the project will advance the integration of AI, optimization, and HCI in robotics, establishing a new paradigm for user-centered robot design and contributing to the growth of AI-driven robotic design automation technologies.

6. References

DoangJoo Alan Synn, Zhifan Guo, Sehoon Ha, HyunJoo Oh, “MotionSmith: A Sketch-Based Design System for Automata Making,” ACM Conference on Human Factors in Computing Systems (CHI), 2026.